

Financial Market Regime Detection Using Hidden Markov Models and XGBoost

Project Overview

This project develops a machine learning framework for automatically identifying financial market regimes from historical price and volume data. The objective is to classify daily market conditions into four economically meaningful states:

- Bull Market
- Bear Market
- Sideways Market
- High-Volatility Market

Rather than relying solely on price trends, the framework combines probabilistic unsupervised learning with supervised machine learning to characterize evolving market dynamics. The resulting regime classifications are incorporated into a confidence-weighted trading strategy whose performance is evaluated against a passive buy-and-hold benchmark.

Objectives

The primary objectives of this project are to:

- Detect hidden market regimes from historical market behavior.
 - Engineer informative technical and market microstructure features.
 - Train a supervised classifier capable of predicting market regimes.
 - Quantify prediction confidence and transition uncertainty.
 - Evaluate strategy performance using risk-adjusted performance metrics.
 - Assess robustness during major financial stress events.
 - Optimize model hyperparameters to maximize trading performance.
-

Dataset

Historical daily price data for the SPDR S&P 500 ETF (SPY) were downloaded using the Yahoo Finance API.

The dataset includes:

- Open

- High
- Low
- Close
- Volume

covering approximately twenty years of market history.

Feature Engineering

A comprehensive feature set was constructed to capture multiple dimensions of market behavior.

Return Features

- Daily returns
- Log returns
- Cumulative returns

Volatility Features

- 10-day rolling volatility
- 20-day rolling volatility
- 60-day rolling volatility
- Average True Range (ATR)

Trend Features

- 20-day moving average
- 50-day moving average
- 200-day moving average
- Distance from each moving average

Momentum Features

- 5-day momentum
- 20-day momentum
- 60-day momentum

Market Microstructure Features

- Daily trading range
- Volume Z-score
- Price efficiency ratio

All features were standardized prior to Hidden Markov Model training.

Hidden Markov Model

A four-state Gaussian Hidden Markov Model was trained to discover latent market regimes from historical observations.

The HMM was configured with:

- Four hidden states
- Full covariance matrices
- Expectation-Maximization optimization
- Maximum of 1000 training iterations

Each discovered state was interpreted using average returns and average volatility to assign financial labels corresponding to Bull, Bear, Sideways, and High-Volatility markets.

Supervised Learning

The HMM-generated regime labels served as targets for an XGBoost multiclass classifier.

Model inputs included engineered features describing:

- returns
- volatility
- momentum
- trend
- trading activity
- market efficiency

A chronological train/test split preserved temporal ordering and prevented look-ahead bias.

Walk-Forward Validation

Model robustness was evaluated using TimeSeriesSplit cross-validation.

For each fold:

1. Train the XGBoost classifier.

2. Predict market regimes for the subsequent time period.
3. Compute classification metrics.
4. Measure strategy performance.

This methodology closely resembles real-world deployment in which future observations are unavailable during training.

Classification Performance

Model performance was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

These metrics quantify the classifier's ability to distinguish among the four market regimes.

Prediction Confidence

Class probabilities generated by XGBoost were used to evaluate prediction confidence.

Additional analyses included:

- Confidence distribution
- High-confidence prediction accuracy
- Confidence spread
- Entropy-based uncertainty estimation

These measures identify periods during which the classifier is highly certain or uncertain about prevailing market conditions.

Regime Transition Analysis

The project investigated how frequently markets transition between regimes.

Analyses included:

- Transition events
- Rolling transition rate
- Transition count statistics
- Transition probability matrix

This provides insight into the persistence and dynamics of different market environments.

Trading Strategy

Predicted regimes were converted into trading positions using a confidence-weighted allocation:

- Bull → Long (100%)
- Bear → Short (100%)
- Sideways → Partial Long (30%)
- High Volatility → Cash (0%)

Positions were shifted by one trading day to eliminate look-ahead bias.

Daily strategy returns were compared with passive market returns.

Performance Evaluation

The strategy was evaluated using several financial metrics:

- Cumulative return
- Sharpe ratio
- Maximum drawdown
- Equity curve
- Confidence analysis
- Stress-period performance

These metrics assess both profitability and risk.

Stress Testing

Strategy robustness was evaluated during major periods of market disruption:

- Global Financial Crisis (2008–2009)
- COVID-19 Crash (2020)

- Inflation and Rate-Hike Bear Market (2022)

Performance was compared directly against a buy-and-hold investment.

Hyperparameter Optimization

A randomized search explored combinations of:

- Tree depth
- Learning rate
- Number of trees
- Row subsampling
- Column subsampling

Each candidate model was evaluated using a custom objective combining:

- Sharpe ratio
- Maximum drawdown

The highest-scoring parameter set was selected for the final model.

Results

The completed framework demonstrates how probabilistic regime discovery can be integrated with supervised machine learning to build adaptive investment strategies. The project combines unsupervised learning, feature engineering, time-series validation, risk-adjusted portfolio evaluation, and hyperparameter optimization into a unified quantitative research pipeline representative of modern systematic trading workflows.

Technologies

- Python
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- XGBoost
- hmmlearn

- yfinance

Skills Demonstrated

- Quantitative Finance
- Machine Learning
- Hidden Markov Models
- XGBoost
- Feature Engineering
- Time-Series Forecasting
- Time-Series Cross Validation
- Statistical Modeling
- Financial Data Analysis
- Risk Modeling
- Strategy Backtesting
- Hyperparameter Optimization
- Performance Attribution
- Python Data Science